

Product Requirements Document:

PawTrace Smart Pet Vest

1. Product Overview

Product name: PawTrace

Product type: Smart pet vest + mobile app

Target users: Urban cat and dog owners, especially busy owners who want to protect their pets from getting lost and notice health problems earlier.

Core value: PawTrace helps owners know where their pet is, identify the pet quickly if it gets lost, and monitor basic health signs in daily life.

2. User Stories and Linked Features

User Story	Linked Features
As a pet owner, I want to see my pet's real-time location on a map, so that I can find my pet quickly if it moves away.	GPS tracking; real-time map display; movement history
As a pet owner, I want to set a safe area for my pet, so that I can receive an alert when my pet leaves that area.	Electronic fence; safe-zone setup; push notification alert
As a pet owner, I want other people to access my pet's identity information easily, so that they can contact me if they find my lost pet.	NFC pet identity card; owner contact page; pet basic information page
As a pet owner, I want to check my pet's heart rate, body temperature, and activity level, so that I can notice possible health problems earlier.	Health sensor data collection; health dashboard; abnormal status reminder
As a pet owner, I want to upload photos of visible symptoms, so that I can get a basic disease-risk suggestion before deciding whether to visit a vet.	Photo upload; AI-assisted health check; risk suggestion result
As a pet owner, I want the app interface to be simple and clear, so that I can use the main functions quickly in daily situations.	Simple mobile app homepage; clear navigation; map, pet profile, and health sections

3. Feature List

Must-have features

1. **Real-time location**

- Show pet location on the app map.
- Show basic movement history.
- Help the owner navigate to the pet.

2. **Electronic fence**

- Allow the owner to draw or select a safe area.
- Send an alert when the pet leaves the safe area.

3. **NFC pet identity card**

- Store pet name, basic traits, vaccination information, and owner contact details.
- Allow a finder to open the information without logging in.

4. **Health monitoring**

- Track heart rate, body temperature, and activity level.
- Show the data in a simple dashboard.
- Mark abnormal data clearly.

5. **AI-assisted health check**

- Allow users to upload a pet photo.
- Provide a basic disease-risk suggestion.
- Remind users that the result is only a reference, not a medical diagnosis.

4. Initial Technical Architecture

Front-end

Mobile app

Main pages:

- Map page: real-time location, electronic fence, alert display
- Pet profile page: pet identity, NFC information, owner contact
- Health page: heart rate, temperature, activity level
- AI check page: photo upload and result display

NFC web page

- Opens when a finder taps the vest with a phone.

- Shows pet identity and owner contact information.
- Does not require login.

Back-end

Core services

- User account and pet profile management
- GPS data receiving and processing
- Electronic fence alert logic
- Health data processing
- NFC identity page generation
- AI photo-check service connection

Data Storage

Database

- User information
- Pet profile
- Owner contact details
- GPS location history
- Safe-zone settings
- Health monitoring records
- AI health check records

File storage

- Uploaded pet photos
- Pet profile images

5. Success Criteria

The first version of PawTrace should be successful if users can:

- Find their pet's location clearly on the map.
- Set an electronic fence without confusion.
- Receive a clear alert when the pet leaves the safe area.
- Open the NFC identity card quickly without login.
- Understand basic health data such as heart rate, body temperature, and activity level.
- Upload a photo and receive a clear AI-assisted health suggestion.